Shenzhen Dudumiao Technology Co., Ltd.

# smokefire Fire, Smoke & Smoking Detection

## Deployment Guide

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Applicable software version: 1.0.0
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# Contents

**Important**
smokefire is a video-AI assistance and alerting tool. It is not a certified fire alarm device and does not replace statutory fire protection, inspections, emergency procedures, or human verification. False positives and missed detections remain possible.

# Document information

| Item | Value |
|---|---|
| Audience | Deployment engineers, administrators, and site operators |
| Delivery | Source-free offline Docker bundles for CPU and NVIDIA GPU |
| Hosts | Windows 10/11, Windows Server, mainstream x86_64 Linux |
| Video input | Direct RTSP, bundled go2rtc, or bulk sync from an existing go2rtc |
| Default URL | http://127.0.0.1:8600 |
| Project | https://github.com/newtv-ai/smokefire |

# 1. Delivery and requirements

## 1.1 Release assets

GitHub Release `v1.0.0` provides two source-free offline deliveries:

- `smokefire-deploy-1.0.0-cpu.zip`: complete CPU bundle;
- `smokefire-deploy-1.0.0-gpu.zip`: GPU configuration, models, scripts, and manuals;
- `smokefire-images.tar.gz.partNN`: all GPU image parts, required with the GPU bundle;
- `SHA256SUMS-v1.0.0.txt`: checksums for Release assets.

Each delivery includes prebuilt smokefire and go2rtc Docker images, `fire_smoke_v5.pt`, `smoking_v4.pt`, Compose files, Windows/Linux scripts, and all six Chinese/English PDFs. The target host does not need Python, Git, compilers, or GitHub access after installation.

## 1.2 Host requirements

| Item | CPU bundle | GPU bundle |
|---|---|---|
| CPU | x86_64, 4+ cores recommended | x86_64, 4+ cores recommended |
| RAM | 8 GB minimum, 16 GB recommended | 16 GB minimum |
| GPU | Not required | NVIDIA GPU, 8 GB+ VRAM recommended |
| Driver | - | Host driver must support the bundled CUDA 12.8 runtime |
| Docker | Engine/Desktop 24+ and Compose v2 | Same, plus NVIDIA Container Toolkit |
| Free disk | 20 GB plus event data | 50 GB plus event data |

| Item | CPU bundle | GPU bundle |
| --- | --- | --- |
| Cameras | RTSP; H.264 recommended | Same |

The CPU build suits pilots and smaller sites. GPU compatibility and capacity are not promised by reference to any single validation-host model. First confirm that the target driver can run the bundled CUDA 12.8 runtime, then validate startup, both model loads, VRAM use, sustained operation, and the planned stream count with real feeds. Actual capacity depends on the GPU, driver, codec, resolution, sampling interval, recording, and network conditions.

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# 2. Download and verification

The repository is currently Public. Users can open the repository and download Release `v1.0.0` without signing in to GitHub:

```
https://github.com/newtv-ai/smokefire
```

## 2.1 CPU bundle

Download and extract `smokefire-deploy-1.0.0-cpu.zip`.

## 2.2 GPU bundle

Download `smokefire-deploy-1.0.0-gpu.zip` and every GPU image part. Extract the bundle and place the parts under `images/`:

```
smokefire/
  images/
    smokefire-images.tar.gz.part01
    smokefire-images.tar.gz.part02
    ...
```

The start script assembles the parts in filename order, loads the Docker images, and removes only its temporary assembled file. Original parts remain intact.

## 2.3 Verify files

Windows:

```
powershell -ExecutionPolicy Bypass -File .\verify.ps1
```

Linux:

```
chmod +x verify.sh start.sh stop.sh configure-go2rtc.sh
./verify.sh
```

Fixed model checksums:

| File | Size | SHA-256 |
|---|---|---|
| models/fire_smoke_v5.pt | 44,014,233 bytes | b5248d55c5d90e341bc4e537887d4bce4b9af1bd0330ddd87825294750a1e216 |
| models/smoking_v4.pt | 44,025,689 bytes | 0ea7b66c2105f1898f56f828b18ffb01934fb1f47a795192e8867ae8fcb128dd |

Do not continue after any checksum failure. Download the affected file again.

# 3. CPU/GPU installation

## 3.1 Windows

Install and start Docker Desktop using Linux containers. The GPU build also requires a working NVIDIA driver and Docker GPU support.

```
docker version
docker compose version
```

After the offline image has been imported by the start script, validate the GPU build with:

```
nvidia-smi
docker run --rm --gpus all smokefire:1.0.0-gpu python -c "import torch;
  print(torch.cuda.is_available(), torch.cuda.get_device_name(0))"
```

Start the service:

```
powershell -ExecutionPolicy Bypass -File .\start.ps1
```

## 3.2 Linux

Install Docker Engine and Compose v2. For the GPU build, also install the NVIDIA driver and NVIDIA Container Toolkit.

```
docker version
docker compose version
nvidia-smi          # GPU bundle only
chmod +x verify.sh start.sh stop.sh configure-go2rtc.sh
./start.sh
```

The script verifies the delivery, selects the CPU or GPU image from `.env`, loads smokefire and go2rtc offline, starts the containers, and waits for `/api/health/ready`.

## 3.3 Lifecycle and logs

```
docker compose ps
docker compose logs -f smokefire
```

Use `stop.ps1` on Windows or `./stop.sh` on Linux. Stopping preserves business data. Never run `docker compose down -v`, which removes named volumes.

# 4. Video integration modes

Choose one mode for the site:

| Mode | Best fit | Configuration effort | Data path |
|---|---|---|---|
| A. Direct RTSP | A few cameras and no go2rtc | One URL per camera | Camera/NVR -> smokefire |
| B. Bundled go2rtc | Cameras are managed in smokefire; detection and recording should share one producer | One URL per camera in smokefire | Camera -> bundled go2rtc -> detection/recording |
| C. Existing go2rtc | The customer already manages feeds in go2rtc | One API base plus one RTSP base | Existing go2rtc -> automatic import -> smokefire |

For an existing go2rtc site, use mode C. Do not copy every camera RTSP URL into smokefire.

The provided configuration script changes only `.env` in the deployment directory. If the service is already running, it recreates the smokefire container while preserving business data.

# 5. Existing go2rtc integration

## 5.1 How the upstream project integrates

smokefire retains the upstream system's bulk synchronization approach:

```
Customer go2rtc GET /api/streams
        |
        v
smokefire synchronizes main-stream names and states
        |
        v
rtsp://customer-go2rtc:8554/<URL-encoded-stream-name>
        |
        v
AI detection, recording, preview, and events
```

The synchronizer only reads `GET /api/streams`; it never calls create, update, or delete operations on the customer's go2rtc. The first sync registers all main streams automatically. When a matching main stream exists, names ending in `_sub`, `-sub`, or `_sd` are treated as substreams and skipped to prevent duplicates. At the default 300-second interval, new streams are added, missing streams are disabled, and returning streams are enabled again.

## 5.2 Configure once

If go2rtc runs on the same Docker host, do not use `127.0.0.1` from inside the smokefire container. Use `host.docker.internal`:

```
.\configure-go2rtc.ps1 -Mode upstream `
  -Api http://host.docker.internal:1984 `
  -Rtsp rtsp://host.docker.internal:8554
```

Linux:

```
./configure-go2rtc.sh upstream \
  http://host.docker.internal:1984 \
  rtsp://host.docker.internal:8554
```

For go2rtc on another LAN host, use its LAN address, for example:

```
API  http://192.168.10.20:1984
RTSP rtsp://192.168.10.20:8554
```

Equivalent `.env` entries:

```
SMOKEFIRE_GO2RTC_ENABLED=false
SMOKEFIRE_UPSTREAM_GO2RTC_ENABLED=true
SMOKEFIRE_UPSTREAM_GO2RTC_API=http://192.168.10.20:1984
SMOKEFIRE_UPSTREAM_GO2RTC_RTSP=rtsp://192.168.10.20:8554
SMOKEFIRE_UPSTREAM_GO2RTC_SYNC_INTERVAL_SEC=300
```

### 5.3 Validate automatic import

1. Open `http://<go2rtc-host>:1984/api/streams` and confirm it returns the expected feeds.
2. Run the configuration script and start smokefire.
3. Open Camera Management and confirm main streams appear automatically with the Upstream badge.
4. Confirm matching substreams were not duplicated.
5. Validate at least one preview, AI inference result, and event recording.
6. Add a test stream and confirm the next sync imports it; remove and restore it to validate automatic disable/re-enable.

Keep existing go2rtc stream names unique and stable. The synchronizer owns imported source URLs; do not create manual cameras with the same names.

## 6. Bundled go2rtc and direct RTSP

### 6.1 Bundled go2rtc

Use this when the site has no go2rtc but wants detection and recording to share one upstream camera producer:

```
.\configure-go2rtc.ps1 -Mode builtin
```

```
./configure-go2rtc.sh builtin
```

Continue adding camera RTSP URLs in Camera Management. smokefire creates internal `sf-camN` aliases automatically, and both AI and recording consume the bundled go2rtc output.

### 6.2 Direct RTSP

```
.\configure-go2rtc.ps1 -Mode direct
```

```
./configure-go2rtc.sh direct
```

Add camera or NVR RTSP URLs individually in Camera Management. This is the simplest mode, but AI and recording may create separate upstream sessions. Prefer go2rtc as the stream count grows.

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## 7. First start and acceptance

Open `http://127.0.0.1:8600` on the service host. Initial model loading may take several minutes.

| Endpoint | Purpose | Expected |
|---|---|---|
| /api/health/live | Process liveness | HTTP 200 |
| /api/health/ready | Models, scheduler, recording, disk, and upstream sync | HTTP 200 |
| /api/status | Runtime and notification metrics | JSON response |

Minimum acceptance:

1. Live Wall, Event Center, and Camera Management open successfully.
2. Video is connected through the selected mode; an existing-go2rtc deployment proves bulk automatic import.
3. Cameras become Online with correct names, images, and timestamps.
4. Authorized test material exercises both the fire/smoke and smoking models.
5. Event details contain snapshot, class, confidence, time, and recording.
6. Confirm True, Mark False Positive, reconnect behavior, restart, and data persistence work.
7. Run the planned stream count and record throughput, latency, GPU/CPU, disk, and network behavior.

Installation success is not production acceptance. See the Model Capability Test Guide for model boundaries and site test design.

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## 8. Configuration and LAN access

The Web port binds to `127.0.0.1:8600` by default. For LAN access, an engineer must set the bind address and allowed hostnames in `.env`, then restart.

| Setting | Default | Purpose |
| --- | --- | --- |
| SMOKEFIRE_IMAGE | Set by CPU/GPU bundle | Application image |
| SMOKEFIRE_ALLOWED_HOSTS | localhost,127.0.0.1 | Accepted Web Host values |
| SMOKEFIRE_GO2RTC_ENABLED | false | Enables bundled go2rtc fan-out |
| SMOKEFIRE_UPSTREAM_GO2RTC_ENABLED | false | Enables existing-go2rtc synchronization |
| SMOKEFIRE_UPSTREAM_GO2RTC_API | Empty | Existing go2rtc API base |
| SMOKEFIRE_UPSTREAM_GO2RTC_RTSP | Empty | Existing go2rtc RTSP base |
| SMOKEFIRE_INFER_WORKERS | 4 | Inference workers |
| SMOKEFIRE_INFER_BATCH_MAX | 8 | Maximum inference batch |
| SMOKEFIRE_STOP_GRACE_PERIOD | 180s | Graceful shutdown period |

RTSP URLs may contain credentials. Restrict access to the deployment directory, backups, and terminal history, and redact full credentials from screenshots and support tickets.

# 9. Routine operation

```
docker compose ps
docker compose logs --tail 200 smokefire
curl http://127.0.0.1:8600/api/health/ready
```

Daily checks should cover camera status, latest successful upstream sync, preview freshness, event playback, notification queues, and disk high-water state. In existing-go2rtc mode, compare the number of go2rtc main streams with Upstream cameras in smokefire.

# 10. Backup, restore, and upgrade

Business data is stored in the `smokefire-data` named volume. Stop the service, record the version, and copy the entire `/app/data` tree. A complete backup includes camera records, the SQLite database, event videos, snapshots, false-positive feedback, and notification outbox. The customer's existing go2rtc configuration remains the customer's backup responsibility and is not stored in the smokefire volume.

Restore data to the matching software version, then validate health, cameras, historical events, recordings, feedback, and upstream sync.

Before upgrading, verify that the backup is readable. Download and verify the new bundle, stop the old version, start from a new directory, and run minimum acceptance. Roll back to the prior image and backup if validation fails.

# 11. Troubleshooting

| Symptom | First check | Action |
| --- | --- | --- |
| Docker command unavailable | Docker Desktop/Engine | Start the engine and retry `docker version` |
| GPU container unavailable | Driver, Container Toolkit, Docker GPU support | Pass `nvidia-smi` and container CUDA checks first |
| GPU image part missing | `images/`, filenames, SHA-256 | Download all parts without renaming them |
| Readiness remains false | Logs, models, disk, upstream sync | Run `docker compose logs --tail 300 smokefire` |
| Existing go2rtc imports nothing | API URL, container reachability to 1984, `/api/streams` | Use `host.docker.internal` on the same host or the LAN IP remotely |
| Fewer imports than go2rtc names | Main/substream naming | `_sub`, `-sub`, and `_sd` matches are deliberately deduplicated |
| Upstream camera becomes disabled | Feed disappeared or sync failed | Restore the feed; the next successful sync re-enables it |
| Camera keeps reconnecting | RTSP 8554, encoded stream name, source state | Test the same go2rtc RTSP URL from the deployment host |
| Too many or no alerts | Threshold, target pixels, scene interference | Replay representative positive and negative samples; do not lower thresholds blindly |
| Event has no video | Disk, warm-up, source stability | Inspect disk status and FFmpeg logs |

# 12. Handover checklist

## Installation

- □ The complete CPU or GPU offline delivery was selected and verified.
- □ smokefire, go2rtc, and both model checks passed.
- □ Start, stop, restart, and host reboot recovery were tested.
- □ `/api/health/live` and `/api/health/ready` return normally.
- □ Install directory, data volume, backup location, and version 1.0.0 are recorded.
- □ The repository remains Public, and anonymous users can open the repository, Release page, and checksum download.

## Video and business acceptance

- □ Direct RTSP, bundled go2rtc, or existing go2rtc mode is explicitly selected.
- □ Existing-go2rtc sites validated one-time setup, bulk main-stream import, addition, removal, and recovery.
- □ Planned cameras have correct names, time, image, inference, and recording.
- □ Both models were tested with representative positive and negative samples.
- □ Snapshots, recordings, Confirm True, false-positive archive, and feedback export work.

- □ Planned capacity, reconnect behavior, and continuous operation were tested.
- □ Users received the User Operation Manual and Model Capability Test Guide and completed handover training.

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